CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M.Sc. Physic Examination November 2019

PS903 Analytical Technique

Date: 11-11-2019 Day: Monday Time: 1.30 PM To 1:50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

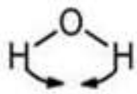
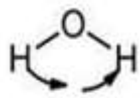
Q-I Choose the correct answer for the following questions. **10**

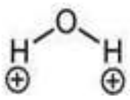
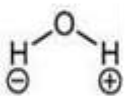
1. Fourier transform, in FTIR, defines a relationship between a signal in ……… domain and its representation in ……. domain 1

(a) time, space (b) space, time

(c) time, frequency (d) frequency, time

2. Choose the correct option for out of plane wagging vibrations of water molecules 1

(a)  (b) 

(c)  (d) 

3. Choose the correct option for Raman spectroscopy method. 1

(a) To study vibrational and rotational modes of the molecules

(b) Useful for structural, crystalline, texture etc. analysis of the sample.

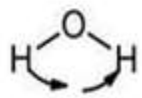
(c) To study elemental analysis of the material
(d) above all

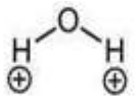
4. Emission of $K_{\alpha2}$ radiation involve the transition of an electron from 1

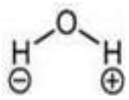
(a) L_i to K shell
(b) L_{ii} to K shell
(c) L_{iii} to K shell
(d) all of above

5. Choose the correct option for out of plane scissoring vibrations of water molecules 1

(a) 

(b) 

(c) 

(d) 

6. How critical current of SQUID loop affected upon application of an external applied flux 1

(a) Increase
(b) Decrease
(c) no change
(d) none of this

7. UV-Vis spectroscopy system relay on which of the following option. 1

(a) Brag-Brentano geometry
(b) Debey-Scherrer's formula
(b) Beer-Lambert law
(d) Photo-electric effect

8. With the help of which of the following equations, the distance calculated from a known wavelength of the source and measured angle? 1

(a) Coolidge equation
(b) Bragg's equation
(c) Debye equation
(d) Scherrer equation

9. X-ray diffractometers are not used to identify the physical properties of which of the following? 1

(a) Metals
(b) Liquids
(c) Polymeric materials
(d) Solids

10. Number of allowed vibrational movements for linear molecule made of N atoms are 1

(a) 3N
(b) 3N-5
(c) 3N-2
(d) 3N-6

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III Semester of M.Sc. Physic Examination November 2019

PS903 Analytical Technique

Date: 11-11-2019 Day: Monday Time: 1.50 PM To 3:30 PM Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I

Q – II Answer the following questions as directed. **10**

1. What is VSM? Where it can be used? 2

2. Define following terms 2

(i) Stokes lines

(ii) Mean free path

3. Name the types of different informations extracted with XRD and SEM. 2

4. Write down Tauc equation and values of n for different allowed/forbidden transitions. 2

5. (I) List two nondestructive technique for material characterization 2

(II) Name two electron source that useful for microscopy techniques

SECTION – II

Q-III Answer the following questions as directed (Any three) **15**

1. Brief the atomic force microscopy method. 5

2. Brief the dynamic light scattering method. 5

3. (I) Find the value of wave number for hydrogen molecule's absorption band for its vibration having frequency 1.5 x 10^{14} vibrations/sec. 5

(II) A cubic crystal was placed in an x-ray diffractometer using incoming x-rays with a wavelength λ=0 .154 nm. A diffraction peak for (110) plane is appeared at 40.3° diffraction angle then calculate the lattice constant of the crystal.

4. Discuss in detail the UV-Visible spectroscopy. 5